

DigiLocker Component Register

Role, responsibilities & privileges for every technical asset — companion sheet to digilocker-architecture-enhanced.svg

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digilocker.yaml

AUTHOR
MODEL DATE
SOURCE MODEL

ID	COMPONENT / TECHNOLOGY	TYPE	ROLE	RESPONSIBILITIES	PRIVILEGES
C-01	Citizen Web Portal <i>web-application · process</i>	AC	Actor-driven client / Citizen-User interface	Renders the citizen-facing web UI; initiates login and consent/document requests; holds the citizen's browser session.	Can submit authentication requests and act on behalf of the logged-in citizen (end-user identity propagation) via the API Gateway. No direct access to any datastore or backend service. No administrative privileges.
C-02	Mobile App <i>mobile-app · process</i>	AC	Actor-driven client / Citizen-User interface	Native mobile UI for citizens; initiates token-based login; requests citizen profile and documents.	Same end-user identity-propagation privileges as the Web Portal via the API Gateway. No direct backend or datastore access.
C-03	API Gateway <i>gateway · process</i>	P	API Gateway / Edge router (DMZ)	Single internet-facing entry point; routes citizen and mobile requests to the Authentication Service and DigiLocker Backend; terminates the public HTTPS connection.	Forwards requests carrying the end-user's propagated identity; invokes the technical-user identity for token validation. No direct datastore access.
C-04	Authentication Service <i>identity-provider · process</i>	P	SSO / Identity Provider	Authenticates citizens (Aadhaar OTP-based login — see digilocker-auth.spthy formal model); issues and validates authentication/session tokens; integrates with UIDAI for eKYC/OTP verification.	Generates authentication-tokens; reads/writes authentication-tokens in the Relational Database using technical-user credentials; calls the external Aadhaar Identity Provider using service credentials.
C-05	Aadhaar Identity Provider <i>identity-provider · external-entity</i>	EE	External Identity Provider (UIDAI / Government, out of DigiLocker trust boundary)	Performs Aadhaar eKYC / OTP verification for citizen identity on request from the Authentication Service.	Provides identity-verification results only. Owned and operated by UIDAI, not DigiLocker; treated as an untrusted external boundary requiring credentialed technical-user authentication.
C-06	DigiLocker Backend <i>application-server · process</i>	P	DigiLocker Platform / core application server	Core business logic for citizen profile, issued/uploaded documents and consent workflows; orchestrates the Document Repository, Relational Database, Consent Manager, Issuer Service, Requester Service, Notification Service and Monitoring Service.	Full read/write access to citizen-profile, issued-documents, uploaded-documents and consent-records via the Document Repository and Database (technical-user credentials). Read-only access to the Issuer Service. Can push verified documents to the Requester Service and emit audit-logs to the Monitoring Service.
C-07	Document Repository <i>database · datastore</i>	DS	Repository / Data store	Stores and serves issued and uploaded documents on behalf of the Backend.	Storage only — no outbound calls. Data encrypted at rest (symmetric shared key, per digilocker.yaml). Accessible only to the DigiLocker Backend, over encrypted JDBC.
C-08	Consent Manager <i>application-server · process</i>	P	Consent Manager	Manages and validates citizen consent for sharing documents with requesting parties; persists consent-records on behalf of the Backend.	Read/write consent-records in the Relational Database using technical-user credentials. No citizen-facing exposure and no access to document content.
C-09	Issuer Service <i>web-service-rest · external-entity</i>	EE	Issuer (Government, out of DigiLocker trust boundary)	Issues government certificates/documents and their digital signatures, retrieved by the Backend.	Supplies issued-documents and digital-signatures to the Backend (read-only from DigiLocker's perspective). External system owned by the Government issuer.
C-10	Requester Service <i>web-service-rest · external-entity</i>	EE	Requester (3rd-party, out of DigiLocker trust boundary)	Requests / consumes citizen documents that DigiLocker shares after consent validation.	Receives issued-documents pushed by the Backend post-consent. Cannot pull data directly or query DigiLocker datastores; external, untrusted boundary.
C-11	Notification Service <i>message-queue · process</i>	P	Notification Service (DMZ)	Sends email/SMS notifications to citizens (e.g. document-shared alerts) on behalf of the Backend.	Receives citizen-profile contact data from the Backend. Outbound-only to citizen communication channels; no datastore access modeled.
C-12	Monitoring Service <i>monitoring · process</i>	P	Monitoring / Audit Service	Collects audit and security events emitted by the Backend; persists audit-logs for security investigation.	Write access to audit-logs in the Relational Database using technical-user credentials. No access to citizen profile, document or consent data.
C-13	Relational Database <i>database · datastore</i>	DS	Repository / Data store	Persists citizen-profile, consent-records, authentication-tokens and audit-logs for the platform.	Storage only. Data encrypted at rest (symmetric shared key). Accessed only by the Authentication Service, Backend, Consent Manager and Monitoring Service, each via encrypted JDBC with technical-user credentials.